01. Photo-electric Effect

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1 Aim of the Experiment

To determine Planck's constant (h) using the photoelectric effect and to study the current-voltage characteristics of the photoelectric tube under varying light intensities and frequencies.

2 Theory

The photoelectric effect is the emission of electrons from the surface of a metal when electromagnetic radiation shines on it. The energy of the emitted electrons is directly proportional to the frequency of the incident light. Below a certain threshold frequency, no electrons are emitted.

According to Einstein's quantum model, light consists of packets of energy called photons. The energy of a photon is given by $E = h\nu$, where h is Planck's constant and ν is the frequency of light. When a photon is absorbed by an electron, part of this energy is used to overcome the work function (W_0) of the material, and the rest becomes the maximum kinetic energy (KE_{max}) of the emitted electron. The equation is given by:

$$h\nu = KE_{max} + W_0 \tag{1}$$

If a negative stopping potential (V) is applied such that even the most energetic electrons cannot reach the collector, the photocurrent becomes zero. At this point, the maximum kinetic energy is equal to the electrical potential energy:

$$KE_{max} = eV \tag{2}$$

Substituting this into Einstein's equation gives:

$$eV = h\nu - W_0 \tag{3}$$

$$V = \frac{h}{e}\nu - \frac{W_0}{e} \tag{4}$$

This is a linear equation. Plotting the stopping potential (V) against frequency (ν) yields a straight line with a slope of h/e. Planck's constant can be calculated from this slope.

3 Procedure

Experiment 1: Measuring Planck's Constant

1. The mercury light source and photodiode enclosures were aligned on a track.
2. The light source and apparatus were allowed to warm up for 10 minutes.
3. The current amplifier was calibrated to zero.
4. Optical filters (365 nm, 405 nm, 436 nm, 546 nm, 577 nm) were placed one by one in front of the photodiode.
5. For each filter, the voltage was adjusted until the photocurrent became exactly zero.
6. This stopping potential was recorded for each wavelength.

Experiment 2: Current Voltage Characteristics (Constant Frequency, Different Intensities)

1. A single filter (e.g., 436 nm) was used.
2. The aperture was set to 2 mm, 4 mm, and 8 mm to change the intensity.
3. For each aperture, the voltage was increased in small steps, and the corresponding photocurrent was recorded.

Experiment 3: Current Voltage Characteristics (Different Frequencies, Constant Intensity)

1. A constant aperture (e.g., 4 mm) was maintained.
2. Different filters (365 nm, 405 nm, 436 nm) were used.
3. For each filter, the voltage was increased in small steps, and the corresponding photocurrent was recorded.

4 Tabular Column

Table 1: Measuring Planck's Constant (V vs f)

Wavelength	**Frequency**	**Stopping Voltage (V)**		
(nm)	**($\times 10^{14}$ Hz)**	**2 mm Aperture**	**4 mm Aperture**	**8 mm Aperture**
365	8.21	−1.920	−1.926	−1.940
405	7.40	−1.543	−1.560	−1.575
436	6.88	−1.357	−1.375	−1.386
546	5.49	−0.819	−0.824	−0.832
577	5.20	−0.700	−0.703	−0.709

Table 1: Stopping voltage measurements for different frequencies and intensities.

Table 2: Constant Frequency, Different Intensities ($\lambda = 436$ nm)

Voltage (V)	Current ($\times 10^{-11}$ A)		
	2 mm Aperture	4 mm Aperture	8 mm Aperture
-4	0	0	0
-2	0	0	0
0	22	62	168
2	73	250	776
4	106	398	1420
6	127	490	1800
8	147	570	2100
10	168	643	2500
12	184	730	2750
14	200	786	3000
16	216	843	3270
18	230	900	3500
20	244	950	3730
22	259	1006	3950
24	272	1060	4180
26	286	1124	4450
28	298	1175	4630
30	306	1225	4780

Table 2: Current-voltage characteristics for varying intensities at constant frequency.

Table 3: Constant Intensity, Different Frequencies (4 mm Aperture)

365 nm		405 nm		436 n	
Voltage (V)	**Current ($\times 10^{-11}$ A)**	**Voltage (V)**	**Current ($\times 10^{-11}$ A)**	**Voltage (V)**	**Cur**
−1.91	30	−0.83	30	−0.95	
−0.95	40	0.11	60	0.14	
0.90	200	0.94	100	0.43	
1.92	340	1.32	120	0.70	
2.90	490	2.12	170	1.01	
3.90	620	3.12	220	1.25	
4.90	710	4.12	250	1.65	
5.92	790	5.12	280	1.82	
7.02	850	6.12	300	1.97	
7.90	900	7.15	330	2.46	
8.82	950	8.16	340	2.98	
9.86	1040	9.13	390	3.97	
10.93	1080	10.13	410	4.96	
11.97	1180	11.17	430	5.97	
12.90	1210	12.10	440	6.97	
13.84	1270	13.10	460	7.97	
14.94	1290	14.12	490	8.96	
15.97	1370	15.12	500	9.95	
16.89	1420	16.14	520	10.92	
		17.11	530	11.92	
		18.14	570	12.92	
		19.25	580	13.92	
				14.93	
				15.93	
				16.93	
				17.93	
				18.93	

Table 3: Current-voltage characteristics for different frequencies at constant intensity (4 mm aperture).

5 Calculations

The photoelectric equation is given by:

$$V = \frac{h}{e}\nu - \frac{W_0}{e} \tag{5}$$

where V is the stopping potential (taken as a positive magnitude for the plot), ν is the frequency of incident light, h is Planck's constant, e is the elementary charge (1.602×10^{-19} C), and W_0 is the work function.

A plot of V versus ν yields a straight line with:

- **Slope** $(m) = \frac{h}{e}$
- **y-intercept** $(c) = -\frac{W_0}{e}$
- **x-intercept** $(\nu_0) = \frac{W_0}{h}$ (Threshold Frequency)

Detailed Calculation for 2 mm Aperture

From the linear fit of the data for the 2 mm aperture, the slope and y-intercept are determined as:

$$\begin{aligned} m_1 &= 3.984 \times 10^{-15} \text{ V} \cdot \text{s} \\ c_1 &= -1.376 \text{ V} \end{aligned}$$

1. Planck's Constant (h_1):

$$\begin{aligned} h_1 &= m_1 \times e \\ h_1 &= (3.984 \times 10^{-15} \text{ V} \cdot \text{s}) \times (1.602 \times 10^{-19} \text{ C}) \\ h_1 &= 6.382 \times 10^{-34} \text{ J} \cdot \text{s} \end{aligned}$$

2. Work Function $(W_{0,1})$: The y-intercept represents $-\frac{W_0}{e}$. Thus, the work function in electron-volts (eV) is the magnitude of the y-intercept:

$$\begin{aligned} W_{0,1} &= -c_1 \times e \\ W_{0,1} &= -(-1.376 \text{ V}) \times 1 \text{ e} \\ W_{0,1} &= 1.376 \text{ eV} \end{aligned}$$

3. Threshold Frequency $(\nu_{0,1})$: The threshold frequency is the x-intercept, where the stopping potential $V = 0$:

$$\begin{aligned} \nu_{0,1} &= -\frac{c_1}{m_1} \\ \nu_{0,1} &= \frac{1.376 \text{ V}}{3.984 \times 10^{-15} \text{ V} \cdot \text{s}} \\ \nu_{0,1} &= 3.454 \times 10^{14} \text{ Hz} \end{aligned}$$

Summary for All Apertures

Applying the same method to the 4 mm and 8 mm apertures yields the following:

For the 4 mm aperture:

- $h_2 = 6.435 \times 10^{-34} \text{ J} \cdot \text{s}$
- $W_{0,2} = 1.384 \text{ eV}$
- $\nu_{0,2} = 3.450 \times 10^{14} \text{ Hz}$

For the 8 mm aperture:

- $h_3 = 6.469 \times 10^{-34} \text{ J} \cdot \text{s}$
- $W_{0,3} = 1.392 \text{ eV}$

- $\nu_{0,3} = 3.446 \times 10^{14}$ Hz

The average values calculated from these measurements are:

$$h_{avg} = \frac{6.382 + 6.435 + 6.469}{3} \times 10^{-34} = 6.428 \times 10^{-34} \text{ J} \cdot \text{s}$$
$$W_{0,avg} = \frac{1.376 + 1.384 + 1.392}{3} = 1.384 \text{ eV}$$
$$\nu_{0,avg} = \frac{3.454 + 3.450 + 3.446}{3} \times 10^{14} = 3.450 \times 10^{14} \text{ Hz}$$

6 Results

The key findings of the experiment are as follows:

- The experimentally determined average value of Planck's constant is $h_{avg} = 6.428 \times 10^{-34}$ J·s.
- The average work function of the photoelectric material is $W_{0,avg} = 1.384$ eV.
- The average threshold frequency is $\nu_{0,avg} = 3.450 \times 10^{14}$ Hz.

The current-voltage characteristics confirm that:

- Saturation current increases with light intensity (aperture size), but the stopping potential remains constant for a given frequency.
- The stopping potential becomes more negative as the frequency of incident light increases.

7 Graph

The following graphs illustrate the data obtained during the experiment:

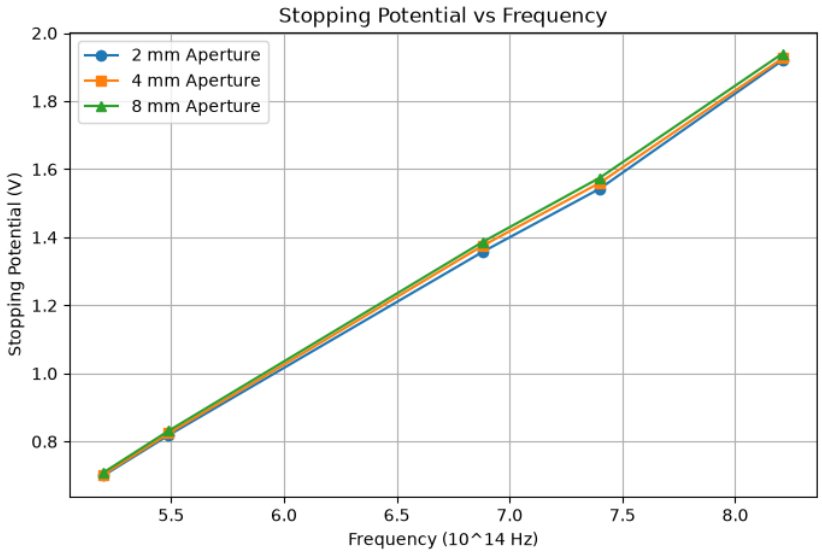


Figure 1: Stopping Potential versus Frequency for different aperture sizes. The slope of these linear fits is used to calculate Planck's constant.

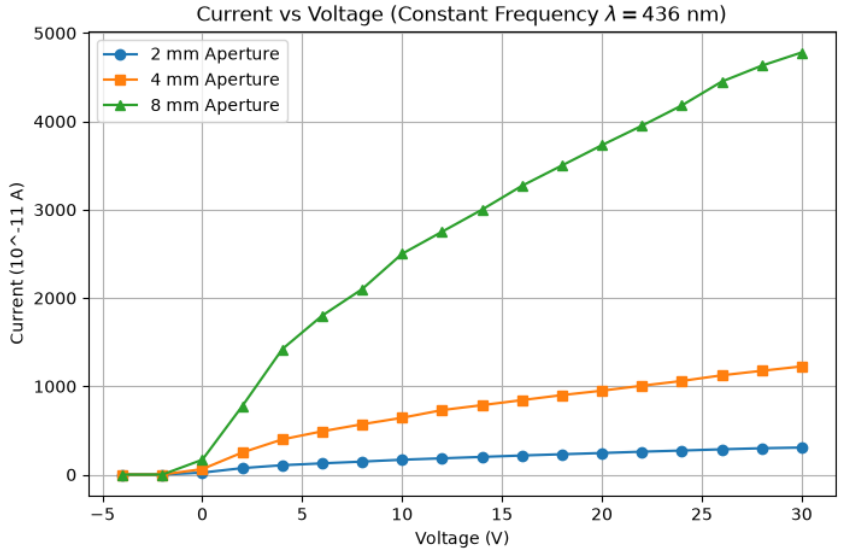


Figure 2: Current versus Voltage for varying light intensities (different apertures) at a constant frequency ($\lambda = 436$ nm). Note how the stopping potential remains constant across intensities.

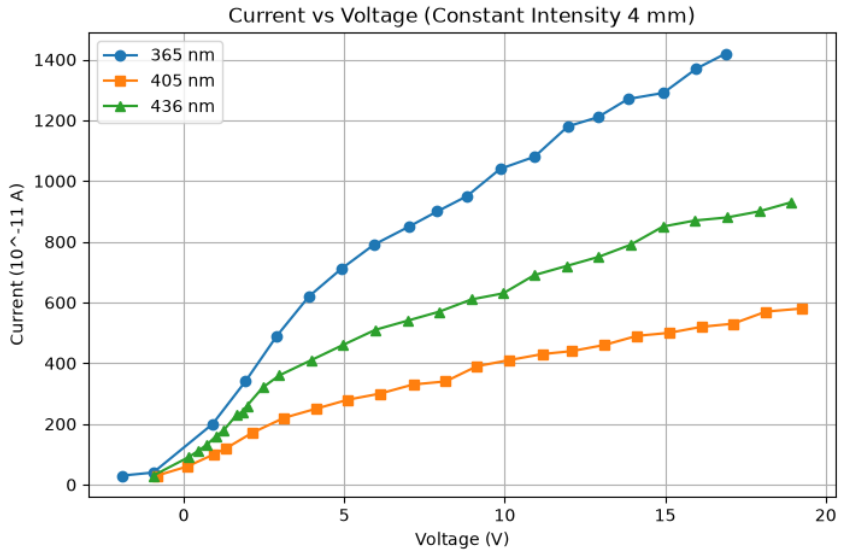


Figure 3: Current versus Voltage for different frequencies at a constant intensity (4 mm aperture). The stopping potential becomes increasingly negative for higher frequencies.

8 Error Calculation

The accepted value of Planck's constant is $h_{accepted} = 6.626 \times 10^{-34}$ J·s.

The percentage error in our experimental calculation is:

$$\begin{aligned}
\text{Percentage Error} &= \left| \frac{h_{experimental} - h_{accepted}}{h_{accepted}} \right| \times 100\% \\
&= \left| \frac{6.428 \times 10^{-34} - 6.626 \times 10^{-34}}{6.626 \times 10^{-34}} \right| \times 100\% \\
&= \left| \frac{-0.198}{6.626} \right| \times 100\% \\
&\approx 2.99\%
\end{aligned}$$

9 Discussion

In this experiment, we observed the fundamental properties of the photoelectric effect. The experiment confirmed that the stopping potential depends on the frequency of the incident light and is independent of its intensity. It also verified that the saturation current depends on the intensity of light but not its frequency. The linear relationship between stopping potential and frequency strongly supports the quantum theory of light.

In real life conditions, small discrepancies between the theoretical assumptions and experimental results can occur due to the reverse anode current and dark current of the cathode. These small currents can make it difficult to pinpoint the exact stopping potential where the true photocurrent is zero. Furthermore, external factors like stray light or instrument calibration could contribute to deviations from the ideal theoretical values, leading to the observed 2.99% error in Planck's constant.

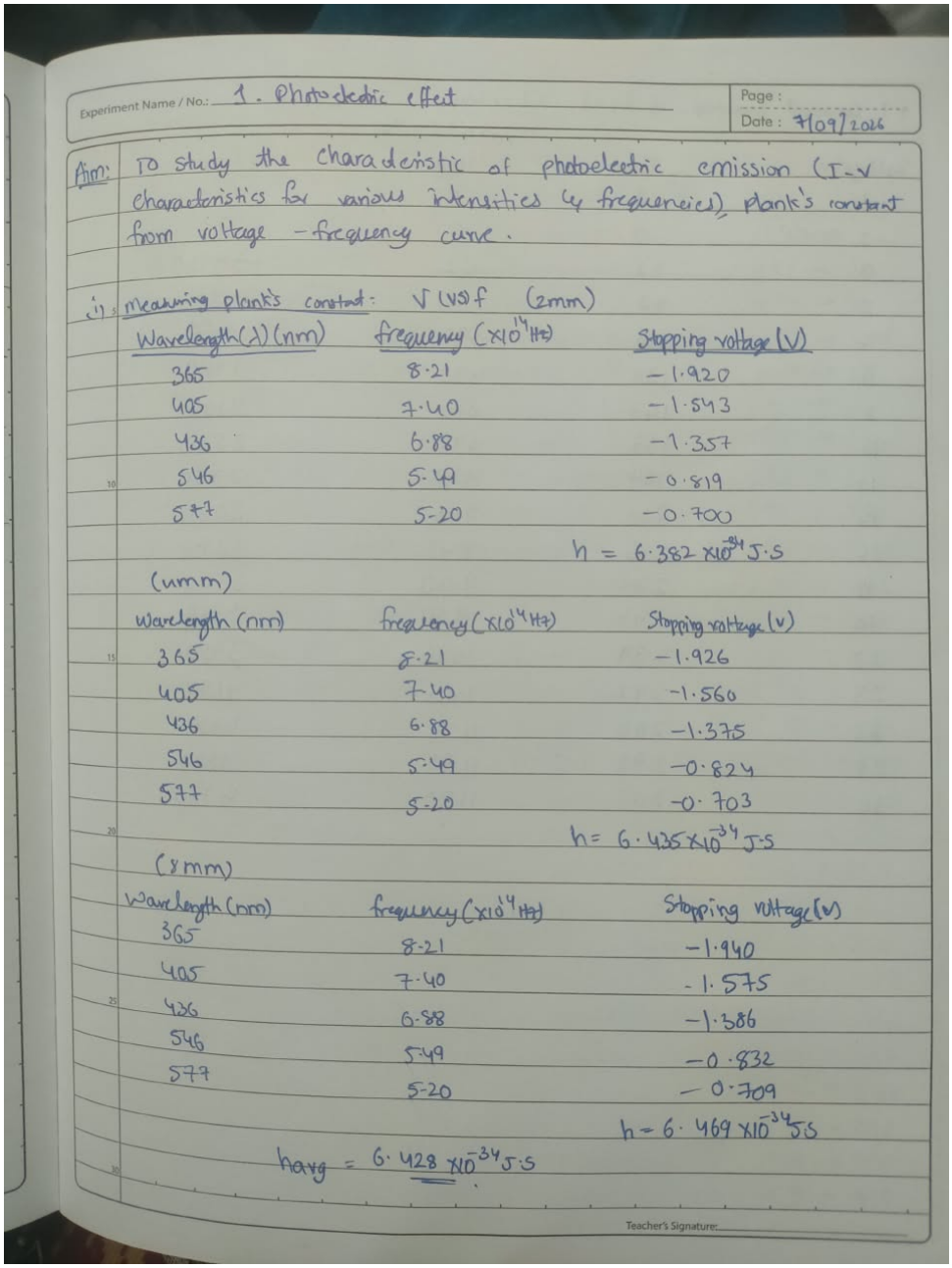
10 References

1. Instruction Manual, Photoelectric Effect Apparatus, Model No. SE 6609, PASCO scientific.

2. Halliday, D., Resnick, R., and Walker, J. Fundamentals of Physics.

11 Lab Observation

The raw observation sheets from the laboratory experiment are attached below.



Experiment Name / No.: 1. Photoelectric effect

Page :

Date : 7/09/2026

Aim: To study the characteristic of photoelectric emission (I-V characteristics for various intensities & frequencies), plank's constant from voltage - frequency curve.

i) Measuring plank's constant: V (V) f (2mm)

Wavelength (λ) (nm)	frequency ($\times 10^{14}$ Hz)	Stopping voltage (V)
365	8.21	−1.920
405	7.40	−1.543
436	6.88	−1.357
546	5.49	−0.819
577	5.20	−0.700

$h = 6.382 \times 10^{-34}$ J·s

(1mm)

Wavelength (nm)	frequency ($\times 10^{14}$ Hz)	Stopping voltage (V)
365	8.21	−1.926
405	7.40	−1.560
436	6.88	−1.375
546	5.49	−0.824
577	5.20	−0.703

$h = 6.435 \times 10^{-34}$ J·s

(8mm)

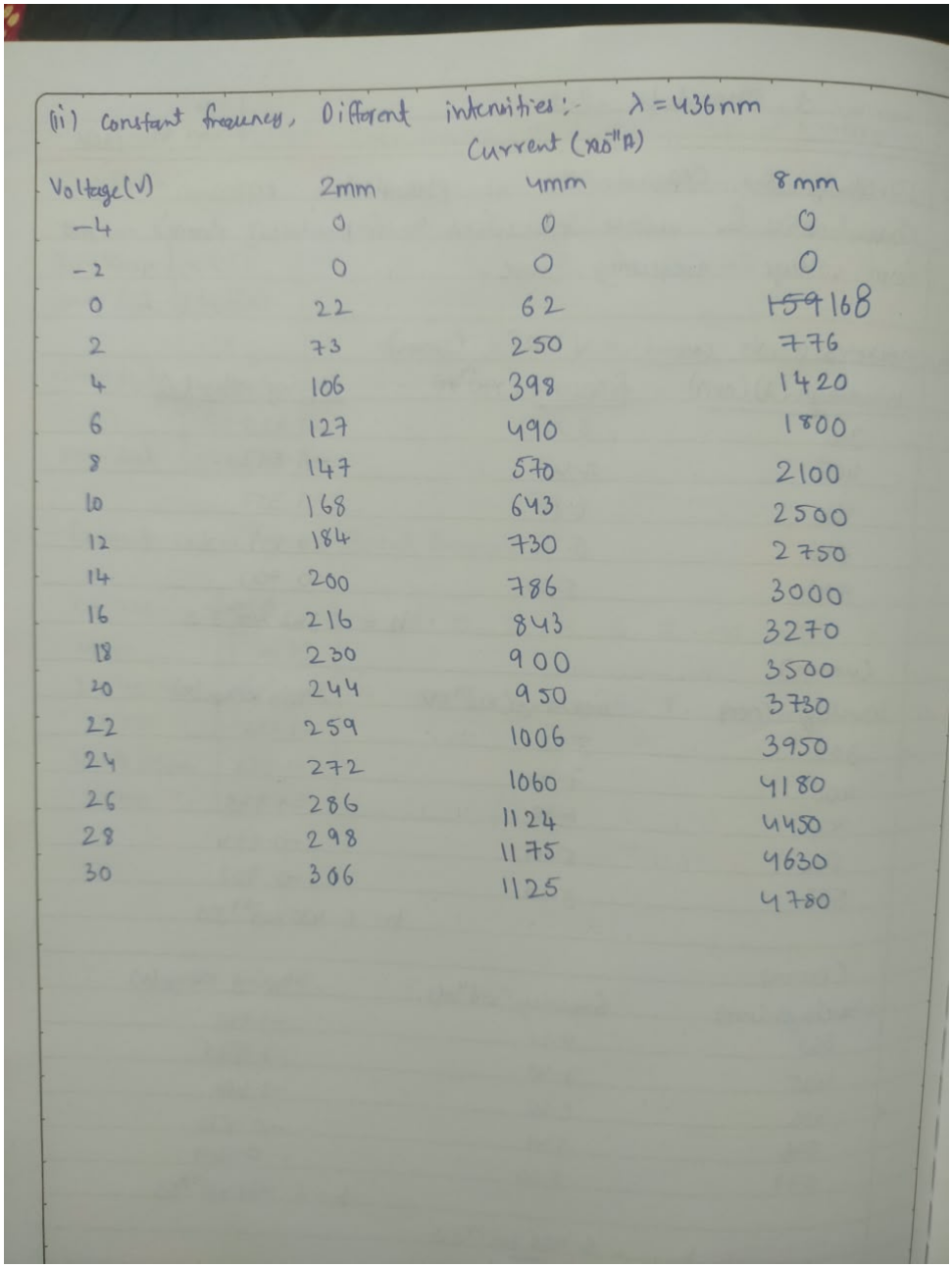
Wavelength (nm)	frequency ($\times 10^{14}$ Hz)	Stopping voltage (V)
365	8.21	−1.940
405	7.40	−1.575
436	6.88	−1.386
546	5.49	−0.832
577	5.20	−0.709

$h = 6.469 \times 10^{-34}$ J·s

$h_{avg} = 6.428 \times 10^{-34}$ J·s

Teacher's Signature:

Figure 4: Raw observation sheet 1: Measuring Planck's constant.

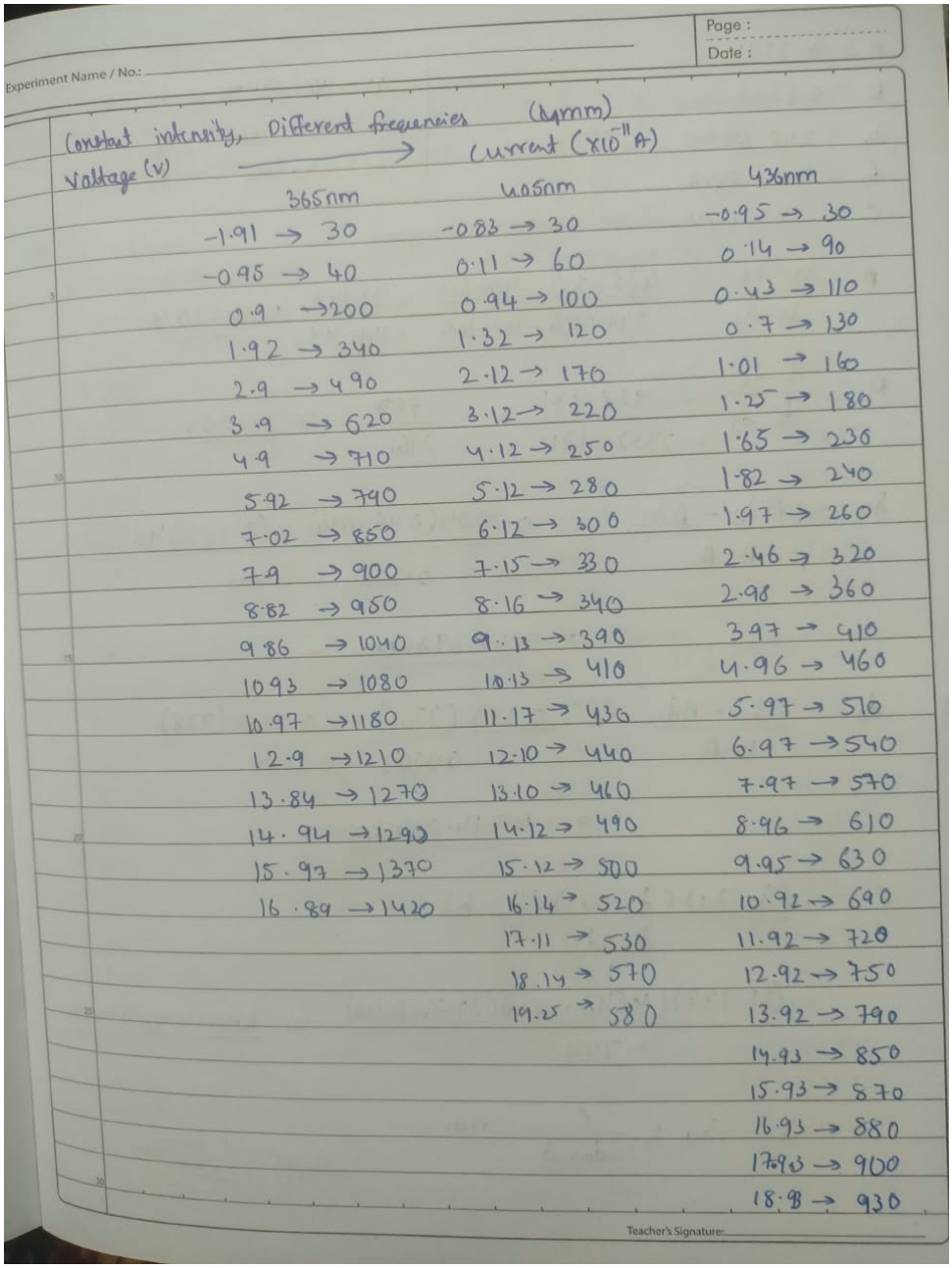


(ii) Constant frequency, Different intensities :- $\lambda = 436\,nm$

Current ($\times 10^{-11}$ A)

Voltage (V)	2mm	4mm	8mm
−4	0	0	0
−2	0	0	0
0	22	62	~~159~~ 168
2	73	250	776
4	106	398	1420
6	127	490	1800
8	147	570	2100
10	168	643	2500
12	184	730	2750
14	200	786	3000
16	216	843	3270
18	230	900	3500
20	244	950	3730
22	259	1006	3950
24	272	1060	4180
26	286	1124	4450
28	298	1175	4630
30	306	1125	4780

Figure 5: Raw observation sheet 2: Constant frequency, different intensities.



Experiment Name / No.:

Page :
Date :

(constant intensity, Different frequencies (4mm)

Voltage (V) → Current ($\times 10^{-11}$ A)

365nm	405nm	436nm
-1.91 → 30	-0.83 → 30	-0.95 → 30
-0.95 → 40	0.11 → 60	0.14 → 90
0.9 → 200	0.94 → 100	0.43 → 110
1.92 → 340	1.32 → 120	0.7 → 130
2.9 → 490	2.12 → 170	1.01 → 160
3.9 → 620	3.12 → 220	1.25 → 180
4.9 → 710	4.12 → 250	1.65 → 230
5.92 → 790	5.12 → 280	1.82 → 240
7.02 → 850	6.12 → 300	1.97 → 260
7.9 → 900	7.15 → 330	2.46 → 320
8.82 → 950	8.16 → 340	2.98 → 360
9.86 → 1040	9.13 → 390	3.97 → 410
10.93 → 1080	10.13 → 410	4.96 → 460
10.97 → 1180	11.17 → 430	5.97 → 510
12.9 → 1210	12.10 → 440	6.97 → 540
13.84 → 1270	13.10 → 460	7.97 → 570
14.94 → 1290	14.12 → 490	8.96 → 610
15.97 → 1370	15.12 → 500	9.95 → 630
16.89 → 1420	16.14 → 520	10.92 → 690
	17.11 → 530	11.92 → 720
	18.14 → 570	12.92 → 750
	19.25 → 580	13.92 → 790
		14.93 → 850
		15.93 → 870
		16.93 → 880
		17.93 → 900
		18.93 → 930

Teacher's Signature:

Figure 6: Raw observation sheet 3: Constant intensity, different frequencies.